

Light Weight Cryptographic Algorithms for Medical Internet of Things (IoT) - A Review

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Abstract: - Internet of things (IoT) is communication between smart objects and human. It finds enormous applications in the field of healthcare monitoring, information management system, agriculture, predicting the natural disaster etc. In all those applications of IoT, security plays a vital role. In this paper, a study on various encryption light weight techniques used for IoT was analyzed. Also the performance of those existing algorithms is analyzed in detail based on the constraint and merits of the IoT. This detailed survey is made towards obtaining a tradeoff between security, cost and performance of IoT based application.

Keywords — Light weight cryptography, IoT, FPGA, Attacks, Throughput

1. INTRODUCTION

Internet of things (IoT) is the communication between machine to machine and machine to human. Here smart object can be a computer/Machine/sensor/ RFID tags. The major application of IoT includes medical crisis supervision and also implemented in transportation, monitoring and managing. It is also play a vital role in Environmental studies which include the wild life monitoring, quality of top soil and water checking. Using IoT, precautionary action can be taken based on the environment changes. It has quite good application in automatic design and building automation. In IoT, security plays a vital role. The services provided by incorporating security over IoT were confidentiality, integrity, authentication and non-repudiation. Many security algorithms were presented in the literature. RC2, RC4, RC6, Advanced Encryption system (AES), Data Encryption System (DES), 3DES, BLOWFISH, TWOFISH, CAST-5, CAST-128, CAST-256, SEED, International Data Encryption standard (IDEA), Tiny Encryption Algorithm (TEA), Extended TEA (XTEA), DEAL, FROG, HEIGHT, TWINE-80, TWINE-128, L-BLOCK, KATAN, PICOLLO, Scalable Encryption Algorithm (SEA), ICEBERG, TWIS, CRYPTION, LED. DSA, YAK, Diffie Hellman Elgamal, Merkle's puzzles Elliptic curve cryptography (ECC). Then few Hash function are listed as follow: MD2, MD4, MD5, SHA -0, SHA-1, SHA-3, GOST, HAVAL, RIPMED-160, TIGER, WHIRLPOO. [19] - [61].

On detailed analysis of various algorithms, most of the mentioned algorithm in the literature is not be applicable for IoT and come under the category of heavy weight or middle weight. The main drawback of heavy weight and middle weight are more number of gates used for implementation. To overcome this issue ISO/ IEC 29192 standards was introduced for light weight cryptographic (LWC) techniques. LWC can be implemented with minimum number of gates and also provide minimum latency and reduce energy consumption. Few of the light weight algorithm are SIMON, SPECK-32, SPECK-64, PRESENT, PRINCE, ROBIN, RECTANGLE, FANTOMAS and Salsa 20. In this paper a detailed survey on various schemes of cryptography namely heavy weight, middle weight and light weight is made in order to propose a novel light weight cryptographic algorithm for IoT applications.

This paper is organized as follows: Section II gives the literature review on the existing cryptographic techniques. A comparative study on various existing technique is made in section III. The conclusion and future work is discussed in section IV.

II LITERATURE REVIEW

Amer Abbas et.al (2014) implemented the VHDL design of PRINCE algorithm on Field programmable gate array (FPGA). The input size of 64 bit and key size of 128 bit followed for the 12 rounds. The steps involved in the algorithm are Round function, Round dependent constant, S- box Layer, Linear diffusion Layer and Middle Involution. The author implemented the proposed PRINCE algorithm in vertex 4 and vertex 6 FPGA kit and they proved vertex 6 gives high throughput, high efficiency and low power consumption. They also proved PRINCE algorithm gives better performance compared to the already proposed ICEBERG and SEA algorithm [1].

Nescimento et.al (2015) implemented block cipher HIGHT (Highest security and light weight) algorithm on FPGA. The block size of 64 bits and key size of 128 bits. It performs three operation namely exclusive OR, modular addition and shift operation. The steps involved in the algorithm are high encryption, key schedule, whitening key generation, constant generation, sub key generation, initial

transformation, round function and final transformation. The author simulated HEIGHT algorithm using Altera Quartus II version 11 and version 13. They proved version 13 gives low latency and high speed. They also proved that Hardware showed better efficiency than software [2].

Sridevi et.al (2015) efficiently implemented the advanced Encryption system (AES) on FPGA. AES performs in four modes. The modes are listed as follows, Electronic code block (ECB), Cipher block chaining, Cipher feedback mode (CFB) and output feedback mode. The input size of 128 bit and variable key size are 128, 192, 256 bits Depends on the size of the key, no of rounds (10, 12, 14) are varied. The steps involved in the algorithm are sub bytes, shift rows, mix column, Add round key. They analyzed the AES algorithm in four ways to increase the throughput as follows: AES was implemented in silicon platform, The structure is modified to pipelined and implemented on FPGA kit, S-box of AES algorithm is replaced by T-box, T-box AES is modified to pipelined architecture. They proved pipelined structure of T-box AES on ECB mode showing higher throughput on FPGA kit [3].

Sohil et.al (2014) implemented the VHDL design of SIMON algorithm was implemented in Xilinx software and Vertex-XC5VFX2007 FPGA kit. Simon algorithm can be implemented for variable Block size are 32, 48, 64, 96, 128 and corresponding key size are 64, 72/96, 96/128, 96/44, 128/192/256. The variable numbers of rounds were followed as follows: 32, 36, 42/44, 52/54, 68/69/72. It performs three operations XOR, AND, left / right shift. They proved Simon algorithm give better performance in hardware when compared with already existed algorithm X-TEA and Humming [4].

Swagata Roy et.al (2014) proposed pipelined architecture for blowfish. The plain text size is 64 bit and key size is 32 bit. Blowfish followed 16 rounds to improve the security. It has a FEISTEL network. This algorithm is synthesized using Verilog HDL and implemented on Spartan 3 FPGA kit (XC35500E-5FG320). They proved that the throughput of non pipelined Blowfish is lesser than the throughput of pipelined Blowfish. They also proved that Blowfish showing better performance than the already existed algorithm AES and DES Algorithm [5].

Hamad et.al (2015) proposed the Pipelined concept of Redundant signed digit (RSD) - Elliptic curve cryptography on Vertex5 FPGA kit. ECC is an asymmetric key algorithm. Generally ECC performs on Elliptic planar curve over the finite field. ECC depends on the point multiplication. Here the point multiplication is a series of point addition and doubling. The size of the elliptic curve determines the difficult of the problem addressed. To reduce the memory size, two operations are followed: They are replacement of carry field arithmetic by the modular arithmetic operation; kartusube-offman method was used. Due to this modification, they achieved the higher throughput, low latency and low memory space [6].

He. and sherli (2015), proposed light weight ECC for internet of things (IoT) to provide better security. They analyzed existing algorithm in three schemes of cryptography. In this they found out that the light weight schemes are suitable for the IoT. Because it consume low power and less memory size [7].

Dhanuka et.al (2015) implemented the ultra light weight Humming bird algorithm on FPGA. The key size of 256 bit and block size of 16 bit, internal state register of 16 bit, linear feedback shift register of 16 bit. The algorithm performs in different Substitution-box as follows: clock dependent s-box, clock independent s-box, resource sharing based s-box, clock dependent 4 s-boxes and clock independent 4 s-boxes. They authors proved that asynchronous design is better than the synchronous design. They also proved 4 s- box implementation proved better result than single s- box implementation. Because it occupy less power and area [8].

Gaurav et.al (2015) implemented the PRESENT algorithm on FPGA. They proved GRP based PRESENT algorithm occupied less memory space when compared to bit permutation based PRESENT algorithm. They also proved that the PRESENT algorithm showing better performance than CLEFIA algorithm. They compared compressed GRP rules PRESENT algorithm, SEA algorithm and ICEBERG algorithm. In this they found compressed GRP rules PRESENT algorithm suitable for IoT [9].

Sufan and Imad (2014) proposed the OLBACA algorithm and implemented on FPGA. OLBACA algorithm can able to withstand Boomerang, Integral and differential attack. The block size of 64 bit, key size of 80 bit and the number of rounds is 22. It performs the operations of key schedule, Permutation, rotation and XOR. They proved that OLBACA occupied less memory space and high processing speed compared to PRESENT algorithm. They also proved OLBACA provide high complexity in 8th round. But, PRESENT provides same complexity in 9th round [10].

Dinu et.al (2015) implemented thirteen algorithms on three different hardware. The Hardware's are 8 bit AT mega microcontroller, 16bit MSP 430 processor and 32 bit ARM processor. The thirteen algorithms are AES, Fantomas, Hight, LBlock, LED, Piccolo, PRESENT, RC5, PRINCE, Robin, SPECK, Simon and Twine. They provided the execution time for encryption and decryption, that including key schedule technique and excluding key scheduling technique [11].

Zhang et.al (2014) implemented RECTANGLE algorithm on FPGA kit. The block size of 64 bits and key size of 80 bits or 128 bits followed for 25 rounds. The steps involved in the algorithm are Add round key, sub column and shift row. They authors proved that RECTANGLE algorithm showing better performance in hardware compared to software. They achieved high efficiency and high throughput by parallel implementation of RECTANGLE algorithm [12].

Gulcan et.al (2015) implemented the Simon algorithm on FPGA for variable block size and key size. They compared the results of Spartan FPGA kit with Vertex FPGA kit. On comparison, they proved that Spartan FPGA achieved better throughput, because it occupied less memory size [13].

Xuanxia yao et.al (2015) Implemented Attribute Based Encryption (ABE) used in Elliptic key Cryptography (ECC). They author proved ABE-ECC is suitable for IoT application [14].

Rezaeian et.al (2014) Proposed 2-slow retiming technique for AES algorithm and implemented on FPGA kit. They proved that throughput can be increased by Pipelining concept, Rolling concept and Sub stage pipelining concept. They also proved that 2-slow retiming technique is better than c-slow retiming technique. The 2-slow retiming technique is breaking the critical path in to number of pieces. Due to that registers are increased. To overcome the more number of register, they author approached multiplication process in third stage of AES algorithm [15].

Poonam jindal et.al (2015) modified the RC6 algorithm as Maximum Column-RC6. The block size of 128 bits and

variable key size of 128/192/256 bits. It is also Feistel structure like Blowfish algorithm. The third stage is similar to the AES algorithm. They author proved that MC-RC6 can overcome the avalanche effect comparable to RC5 and RC6. It provided high throughput on implemented in FPGA kit. They also proved that it will overcome the Linear and differential cryptanalysis attack [16].

III DISCUSSIONS

The Symmetric algorithm is faster than Asymmetric algorithm. The block cipher is slower than the stream cipher. Efficiency and throughput is increased in Hardware compared with software. In Hardware, FPGA shows better results than other hardware like Microcontroller and ARM processor. On FPGA, Vertex 6 board throughput is very higher than the vertex 4 board. In software Altera Quartus II- Version 13 output better than the Version 11. Transfer bit rate is higher in Wi-Fi than Zigbee. Processing time is reduced by the Pipeline concept, sub-stage pipeline concept and rolling concept. The S-P network algorithm is better than S-box algorithm and Feistel network algorithm.

TABLE 1. COMPARISON ON VARIOUS EXISTING SECURITY ALGORITHMS

Author	Algorithm	Problem Addressed	Block Size	Key Size	Rounds	Methodology	Output	Inference
Amer Abbas et.al (2014)[1]	PRINCE	Throughput and Efficiency in hardware	64	128	12	Implemented in FPGA-Vertex 6, Vertex 4	Throughput=4.18Gbps, Efficiency= 8.681 Mbps/Slice	Better then ICEBERG and SEA
Nescimento et.al (2015)[2]	HIGHT	Efficiency in software or Hardware	64	128	32	Implemented in FPGA, Altera Quartus II, Version 13	Transfer rate = 157 Mbps, Delay=9.8ns, F=11.35 MHZ	Hardware is better than software
Sridevi et.al (2015)[3]	AES	Area and throughput in Hardware	128	128	10	Parallelism for third Stage Mix column	Throughput=25.54, Area=30% reduced	Pipelined Concept- Reduced the Area and increase the Speed
Sohil et.al (2014)[4]	SIMON	Number of Gates in Hardware	32,48, 64,96, 128	Variable	32,36, 42,44, 52,54	Implemented In FPGA	32 slices in SPARTAN 3 and 6 FPGA	Better than X-TEA and Humming bird
Swagata roy et.al (2014)[5]	BLOWFISH	Throughput in Hardware	64	32-448	16	Implemented in Spartan 3E XC35-500E-SFG 320	Throughput=6.3 Gbps, F=295.63 MHZ	Better then AES and DES
Hamad et.al (2015)[6]	ECC-RSD	Processing speed	NA	NPK C	NA	RSD, Pipeline KATATUSUBE OFFMAN	LUT=34612, F=160MHZ Time= 2.26 ms	Pipelined architecture increase the Speed
He and Sheli (2015)[7]	ECC	Health care Monitoring	NA	NPK C	NA	Light weight Scheme	Light weight ECC suitable for IoT	Less RAM size
Dhanuka et.al (2015)[8]	HUMMING BIRD	Low power and Low Memory size	16	256	NA	Different S-box method	less power and Area	Asynchronous better then Synchronous Design
Gaurav et.al (2015)[9]	PRESENT	Number of Gates in Hardware	64	128	32	GRP-COMPRESSED	Reduced from 3200 to 2980 GE	Better then CLEFIA and ICEBERG

Sufan et.al (2014)[10]	OLBCA	To Overcome Boomerang, Integral and differential attack	64	80	22	Key Scheduling Technique	1521.06 Gates used	Better then PRESENT
Dinu et.al (2015)[11]	Triathlon(13 algorithm)	Light weight Algorithm	Variable	Variable	Variable	Implemented in ARM, Microcontroller	PRESENT,PRINCE,SIMON, are applicable	Depends upon the application light weight algorithm used
Zhang et.al (2014)[12]	RECTANGLE	Processing speed	NA	80	NA	one cycle per Round	1467 GE ,246K bits/sec throughput	Hardware is better than software
Gulcan et.al (2015)[13]	SIMON	Memory size in Hardware	32	64	32	Implemented Vertex-5 FPGA	F = 523.56 MHZ, LUT=177, Register=182	Hardware is better than software
Xuanxia yao et.al (2015)[14]	ECC	Tradeoff between security and Privacy	NA	NA	NA	Attribute base ECC	Light weight ECC	Provide High Security
Rezaeian et.al (2014)[15]	AES	Number of registers	128	128	10	2-slow retiming Technique	Throughput = 86 Gb/s, F = 671.52 MHZ	2-slow better then c-slow retiming
Poonam et.al (2015)[16]	MC-RC6	Avalanche Effect	128	128, 192, 256	20	MC-Increased number of column	Avalanche effect(AE) = 0.45	AE is lesser then RC5 and RC6

IV CONCLUSION

Due to the advancements in the technology, Internet of Things becomes part in our day to day life. Even though it finds enormous applications, it is lack in security. In this paper, a detailed review on various security algorithms is done. Comparison between the various algorithms is also made in terms of its key size, block size and its performance. This gives an overview on the limitations of the existing security techniques and lays a platform to propose a novel light weight technique with minimum number of block size and key size.

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